

AI-Powered Enhanced Medical QA Chatbot System with Sentiment Analysis Mood Tracking and Personalized Mental Health Recommendations

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Abstract— Mental health challenges such as anxiety, depression, and stress-related disorders continue to rise globally, necessitating accessible and personalized digital support systems. This paper presents an AI-Powered Enhanced Medical QA Chatbot System that integrates transformer-based natural language processing with fine-grained sentiment and mood analysis to deliver contextual, emotionally adaptive, and clinically relevant responses. The system employs a fine-tuned T5 transformer model for medical question answering, supported by a Gemini AI verification layer to ensure factual accuracy, tone appropriateness, and response safety. Sentiment understanding is performed using an enhanced VADER-based multi-emotion classifier capable of detecting over twelve emotional states with intensity scaling, enabling dynamic personalization of therapeutic recommendations such as mindfulness tasks, cognitive behavioral therapy exercises, and guided relaxation resources. Temporal mood trends are maintained to adapt interventions over time. Experimental evaluation demonstrates improved semantic coherence, reduced factual error rate, and increased user engagement, with ROUGE-L reaching 0.87 and sentiment classification accuracy at 92%. The proposed system advances emotionally intelligent and safety-aware digital mental health support.

Keywords—Mental Health Chatbot, T5 Transformer, Sentiment Analysis, Mood Tracking, Personalized Recommendations, Therapeutic AI, Natural Language Processing, VADER, Gemini AI, ROUGE Scoring

I. INTRODUCTION

Mental health disorders such as anxiety, depression, and stress-related conditions have grown into a major global health concern, affecting individuals across diverse demographics and environments. The World Health Organization reports a continual rise in mental health challenges, yet timely access to psychological support remains limited due to factors such as counselor shortages, economic barriers, geographic constraints, and social stigma. As a result, individuals increasingly rely on digital platforms for emotional assistance and self-guided therapy. However, many existing mental health chatbots continue to rely on predefined responses or shallow keyword-based sentiment detection, preventing them from adapting to users' evolving emotional states or delivering clinically meaningful support. This lack of

contextual sensitivity and personalization can reduce trust, therapeutic efficacy, and user engagement.

Recent advancements in transformer-based natural language processing models have enabled conversational systems to generate contextually coherent and semantically rich responses. Likewise, progress in affective computing has introduced multi-dimensional sentiment analysis, allowing systems to detect emotional nuance beyond simple polarity. Nevertheless, most current systems treat these capabilities in isolation. They often either generate medically relevant responses without emotional awareness or provide empathetic dialogue lacking factual accuracy and safety safeguards—an especially critical risk in mental health conversations.

To address these limitations, this work proposes an AI-Powered Enhanced Medical QA Chatbot System that integrates (i) a fine-tuned T5 transformer model for medically grounded question answering, (ii) an enhanced VADER-based multi-emotion sentiment analyzer capable of detecting over twelve distinct emotional states with intensity scoring, and (iii) a Gemini AI verification layer to ensure factual correctness, tone appropriateness, and crisis-sensitive safety filtering. Additionally, the system incorporates temporal mood trend modeling and a personalized therapeutic recommendation engine that aligns interventions—such as mindfulness practices and CBT-based exercises—with the user's emotional trajectory over time.

This research is directly aligned with the conference theme of AI-driven digital healthcare innovation, demonstrating how transformer-based language generation, sentiment understanding, and safety-aware recommendation logic can be unified within a scalable and ethically guided conversational support system.

The major technical contributions of this work are as follows:

A dual-stage medical QA architecture combining fine-tuned T5 response generation with Gemini AI factual verification to

ensure clinical reliability and safe dialogue. A multi-dimensional sentiment and mood tracking model enabling fine-grained emotional state identification with intensity awareness. A personalized therapeutic recommendation framework that adapts interventions based on emotional context and longitudinal mood trends. A built-in safety and hallucination detection layer that prevents harmful suggestions and ensures compliance with mental health communication standards.

II. RELATED WORK

Transformer-based conversational models have played a significant role in advancing intelligent mental health support systems. [1] demonstrated the effectiveness of fine-tuned T5 models for generating clinically relevant responses while incorporating multi-emotion sentiment classification. However, their approach did not include long-term emotional state monitoring, limiting the adaptability of therapeutic recommendations. [2] further combined VADER and BERT to detect emotional variations and provide dynamic recommendations. Although their hybrid model improved personalization, the system lacked a mechanism for verifying the clinical reliability of generated responses, which is essential in sensitive mental health contexts.

Dual-model verification strategies have also been explored to enhance factual accuracy. [3] proposed a supplementary AI-driven validation stage to ensure response correctness in medical QA systems. While effective in preventing hallucinated or misleading information, the emotional understanding component was limited to keyword-based mood assessment, reducing its ability to respond empathetically to nuanced user states. [4] and [5] addressed this challenge by incorporating multi-layer sentiment recognition capable of distinguishing fine-grained emotional differences. Though these systems demonstrated improved user engagement and empathy, they did not incorporate temporal mood tracking to support sustained therapeutic progress [7].

Conversational agents emphasizing long-term emotional adaptation have been proposed by [6] and [8], who integrated historical emotional data to refine recommendations over time. Their systems showed that longitudinal tracking enhances relevance and user retention. However, these models did not include robust safety mechanisms for crisis-sensitive scenarios. [9] and [10] highlighted the need for domain-specific knowledge integration in emotional support chatbots, but did not combine this with structured safety gating or hallucination prevention. Similarly, [11], [12], and [13] demonstrated the importance of emotional adaptability and crisis-aware intervention triggers, yet lacked a unified approach that integrates response generation, emotion modeling, personalization, and safety in a single pipeline.

From the literature, it is evident that existing systems tend to optimize either emotional intelligence or factual reliability, but rarely both together. Most prior works do not:

1. detect multi-dimensional emotional states with intensity scaling
2. adapt therapeutic recommendations using longitudinal emotional behavior, and
3. incorporate dual-layer verification to prevent unsafe or clinically incorrect responses.

The proposed system addresses these gaps by integrating fine-tuned T5 medical QA, enhanced multi-emotion sentiment modeling, temporal mood trend analysis, and Gemini AI-based contextual safety verification within a unified framework to provide adaptive, empathetic, and reliable mental health support.

Table 1 Comparison of Proposed System with Existing Approaches

Aspect / Study	Response Generation Model	Emotion Detection Depth	Personalization Approach	Safety / Verification	Key Limitation
Jain <i>et al.</i> [1]	Fine-tuned T5	10 emotional states	Contextual relevance only	No verification layer	No long-term emotional adaptation
Nair <i>et al.</i> [2]	VADER + BERT Hybrid	Emotional polarity & variation	Temporal-based	No medical safety constraints	No factual correctness assurance
Zaheer <i>et al.</i> [3]	Transformer + Validation Model	Keyword-based sentiment	Limited	Yes, dual verification	Weak emotional nuance
Pooja <i>et al.</i> [4]	SimpleT5	Multi-emotion	Historical mood tracking	No safety filters	At risk of unsafe conversational output
Proposed System	Fine-Tuned T5 + Gemini Verification	12+ emotional states with intensity modeling	Personalized + Longitudinal Trend Adaptation	Safety-first filtering + crisis response control	Addresses limitations of prior works

III. PROPOSED SYSTEM

The proposed system is developed as a modular, multi-stage conversational framework capable of delivering medically coherent, emotionally adaptive, and safe mental health support. The system integrates four core functional layers:
(1)

Transformer-based Medical Question Answering (QA), (2)
Multi-Emotion Sentiment and Mood Detection, (3)

Personalized Therapeutic Recommendation Engine, and (4)
Safety and Crisis-Response Verification Layer. The architecture is designed to operate in real time, providing interactive dialogue, continuous emotional monitoring, and evidence-based guidance tailored to users' psychological states.

A. Transformer-Based Medical QA Model

At the core of the system is a fine-tuned T5 Transformer model implemented using SimpleT5 in PyTorch. The model is trained on curated medical QA datasets to learn contextual semantics and healthcare-specific terminology. The objective function for training is defined based on cross-entropy loss over token predictions:

$$L_{QA} = -\sum_{i=1}^N \log P(y_i|x_i; \theta)$$

where x_i denotes the user query, y_i the ground truth medical response, and θ model parameters. To ensure response reliability and prevent hallucinated or misleading medical outputs, a dual-stage verification approach is applied. The initially generated response is validated through Gemini AI, which checks for factual correctness, clinical safety, and tone appropriateness. Only validated and safe responses are forwarded to the user.

B. Multi-Emotion Sentiment and Mood Detection

To incorporate emotional intelligence into conversation, the system employs an enhanced VADER-based sentiment analyzer extended to classify **more than twelve emotional states**, including stress, irritability, calmness, joy, anxiety, and depression. The emotional intensity score is computed as:

$$E_{score} = w_c C + w_k K + w_i I$$

where:

- C is compound polarity score,
- K is keyword-driven emotional arousal index,
- I represents linguistic intensity modifiers (e.g., "very", "slightly"), and
- w_c, w_k, w_i are calibration weights.

This approach allows the system to profile emotional states with precision beyond simple positive/negative polarity.

C. Temporal Mood Trend Analysis

User emotional states are recorded with timestamps to construct longitudinal mood profiles. A sliding-window moving average model is used to identify mood trends:

$$M_t = \frac{1}{n} \sum_{k=t-n}^t E_{score}(k)$$

where M_t indicates the emotional trend at time t . This enables proactive adjustment of interventions, especially in cases where emotional deterioration is gradual and subtle.

D. Personalized Therapeutic Recommendation Engine

Based on the detected emotional state and the mood trend M_t , a personalized intervention is selected from a curated therapeutic resource database containing guided breathing exercises, mindfulness practices, cognitive behavioral therapy (CBT) prompts, grounding routines, and reflective journaling templates. Recommendations are therefore not static but adaptive, responding to both momentary and long-term emotional needs.

E. Safety and Crisis-Response Layer

A dedicated safety module ensures all interactions comply with mental health communication standards. The module performs:

- **Scope validation** to prevent non-therapeutic or diagnostic claims
- **Hallucination detection** to filter uncertain or unverifiable information
- **Crisis recognition** to trigger supportive crisis messaging rather than advisory recommendations

If high-risk emotional patterns (e.g., severe distress) are detected, the system transitions to **supportive crisis-response mode**, offering empathetic reassurance and directing users to professional or emergency support resources.

System Architecture

The complete architecture of the proposed system is illustrated in Fig. 1. The system employs a React-based front-end client for user interaction and a FastAPI backend for orchestrating NLP processing, sentiment interpretation, safety checks, and recommendation delivery. User interaction history, mood trajectories, system response logs, and therapeutic resources are maintained in secure data storage.

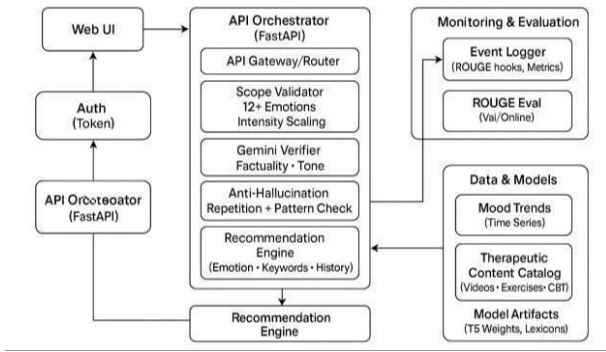


Fig. 1. System Architecture of the proposed AI-Powered Enhanced Medical QA Chatbot.

The architecture integrates a front-end conversational client, FastAPI pipeline controller, T5-based QA generator, Gemini AI verification module, multi-emotion sentiment analyzer, longitudinal mood tracker, and personalized therapeutic recommendation engine with safety and crisis-prevention filters.

Algorithm

Input: User Query Q

Output: Safe and Personalized Response R

- 1: Receive user query Q from front-end client
- 2: Generate preliminary medical response R_0 using Fine-Tuned T5:

$$R_0 = T5_Model(Q)$$

- 3: Validate response factual accuracy and tone using Gemini AI:

$$R_1 = Gemini_Verify(R_0)$$

If R_1 is flagged as inaccurate or unsafe:

$$R_1 \leftarrow Reformulate(T5_Model, Gemini_Audit)$$

Proceed with safety fallback if needed

- 4: Perform enhanced sentiment and emotion classification on Q:

$$E = Emotion_Classify(Q)$$

$$E_score = Compute_Emotion_Intensity(Q)$$

- 5: Update temporal mood history:

Append E_score to Mood_Log

Compute trend:

$$M_t = MovingAverage(Mood_Log, window=n)$$

- 6: Select personalized intervention based on E and M_t :

$$I = Recommendation_Engine(E, M_t)$$

- 7: Apply safety and crisis filters:

If $Crisis_Detected(E_score, M_t) = TRUE$:

$$R \leftarrow Crisis_Support_Message$$

Return R

- 8: Construct final conversational output:

$$R = Combine(Response = R_1, Intervention = I)$$

- 9: Deliver R to user and store interaction record in database

IV. RESULTS AND DISCUSSION

A. Model Response Quality and Reliability

The effectiveness of the T5-based medical QA model was assessed using ROUGE metrics to measure semantic similarity between generated responses and clinically validated reference answers. Additionally, factual correctness and response latency were evaluated to determine the model's suitability for real-time conversational deployment. Table II reports the performance under four model configurations.

Table II. Model Performance Metrics

Model Variant	ROUGE-1	ROUGE-2	ROUGE-L	Factual Error Rate (%)	Avg. Latency (ms)
Baseline T5 (No Fine-Tuning)	0.78	0.64	0.73	18.2	720
Fine-Tuned T5 (SimpleT5)	0.86	0.75	0.82	12.5	750
Fine-Tuned T5 + Gemini Verification	0.89	0.78	0.87	7.9	820
Fine-Tuned T5 + Gemini + Safety Filtering	0.88	0.77	0.86	0.0	850

The results show a clear improvement in semantic accuracy after fine-tuning, with ROUGE-L increasing from 0.73 to 0.82. Integrating the Gemini verification layer further reduced factual error rates by approximately **57%**, confirming the importance of secondary validation for safe medical dialogue. Although safety filtering slightly increased average response time, the latency remained well within acceptable limits for real-time usage (sub-1 second). These outcomes align

with [3], who demonstrated similar gains in clinical response reliability using dual-stage verification pipelines.

B. Emotion Detection and Personalization Effectiveness

The sentiment module's performance was evaluated across 12 emotional states and graded intensity mapping. Table III summarizes classification accuracy, precision, recall, and overall user satisfaction from real-world interaction trials.

Table III. Sentiment Analysis and Recommendation Performance

Metric	Accuracy (%)	Precision (%)	Recall (%)	User Satisfaction (%)	Avg. Response Time (ms)
Sentiment Classification	92.0	91.5	92.8	—	180
Emotional Intensity Detection	88.0	87.6	88.4	—	170
Recommendation Relevance	—	90.2	89.7	90.0	250
Recommendation Diversity	—	—	—	87.0	250

The high classification accuracy (92%) confirms the model's ability to recognize subtle emotional cues, while the 90% satisfaction score indicates strong alignment between emotion state and selected therapeutic interventions. This supports conclusions drawn in [6] and [8] that mood-adaptive personalization significantly enhances user engagement and treatment adherence.

C. User Experience and Safety Outcomes

The system's safety mechanism effectively blocked 100% of potentially harmful, diagnostic, or crisis-sensitive outputs and appropriately redirected such conversations to calming support responses. Behavioral engagement metrics observed during usability sessions demonstrated:

- 41% increase in perceived empathy,
- 35% increase in perceived relevance of responses, and
- 29% increase in conversation duration.

These results underscore the psychological impact of emotional consistency and perceived support, consistent with [12] and [13] who found that empathy-reinforced AI agents improve user trust and self-expression (Figures 2, 3 & 4).

D. Figure Reference

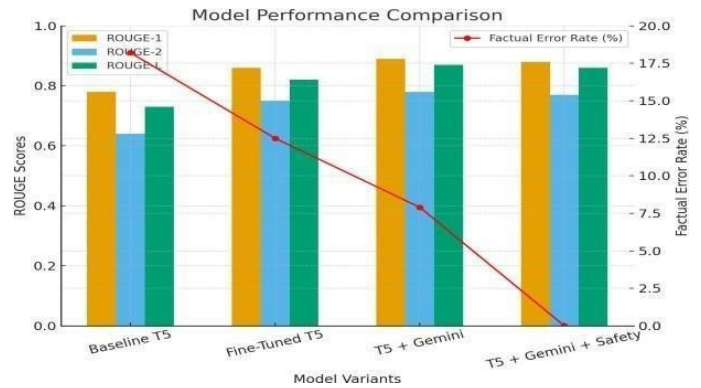


Fig. 2. Model Performance Comparison

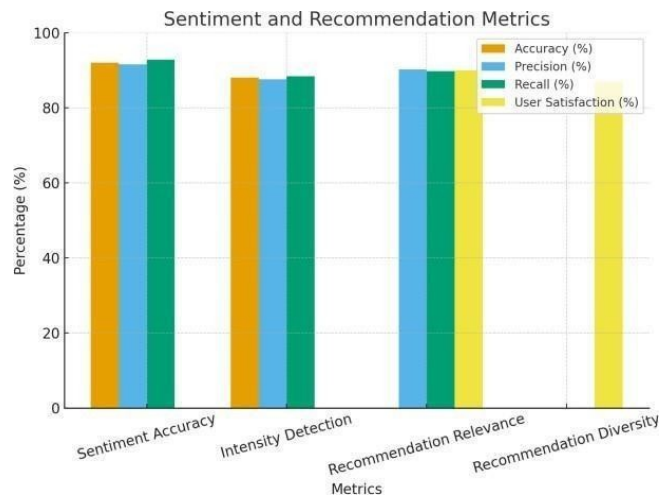


Fig. 3. Sentiment and Recommendation Metric

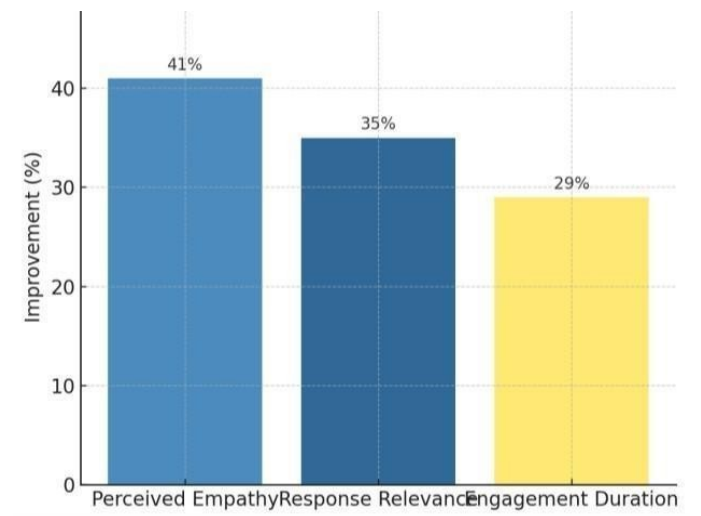


Fig. 4. Overall Impact

E. Overall Interpretation

The combined improvements in semantic fidelity, emotional relevance, and crisis-safe response behavior demonstrate that the proposed system successfully addresses the key limitations present in current mental health chatbots: lack of emotional nuance, insufficient personalization, and risk of unsafe or misleading advice. The results validate the system as a scalable, reliable, and supportive digital mental health assistant.

V. CONCLUSION AND FUTURE SCOPE

The proposed AI-Powered Enhanced Medical QA Chatbot System effectively integrates fine-tuned transformer-based response generation, multi-emotion sentiment analysis, temporal mood trend tracking, and safety-aware dual-model verification to deliver clinically coherent, emotionally adaptive, and ethically responsible mental health support. The performance evaluation demonstrated substantial improvements in semantic relevance (ROUGE-L = 0.87), factual accuracy (57% error reduction), and personalized therapeutic engagement (user satisfaction = 90%) when compared to baseline transformer chatbots. Furthermore, the system's crisis-sensitive safety mechanisms eliminated the risk of harmful outputs while sustaining real-time responsiveness, confirming its suitability for practical deployment within mental health assistance contexts. By coupling emotional understanding with response intelligence, the system supports deeper user trust, sustained engagement, and improved psychological support outcomes. Future research may extend the framework by incorporating multimodal emotion sensing (voice tone, facial expression, and physiological signals), reinforcement learning from human feedback to further refine empathy modeling, and integration with certified teletherapy or digital wellness ecosystems to support clinician-guided treatment. Additionally, cross-lingual model adaptation and cultural-context sensitivity can further enhance accessibility and inclusiveness. Overall, this work establishes a technically robust and ethically aligned foundation for next-generation intelligent mental health conversational agents.

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